# EE5203 L12 Worksheet

*System integration and diagnosis - Basri Erdogan*

**Expected time:** 30-45 minutes **Policy:** Attempt every question before consulting the worked solutions. Several questions ask about **your own project** — answer for it, not in general.

## Part A - Resource conflicts

1. Name the three species of failure and, for each, the tool that finds it.
2. Here is a proposed resource map. Find **three** conflicts and explain each.

| Resource | Pin(s) | Owner |
|---|---|---|
| TIM2 CH1 | PA5 | LED PWM, 1 kHz |
| TIM2 CH2 | PA1 | servo PWM, 50 Hz |
| ADC1 IN1 | PA1 | light sensor |
| SPI1 SCK | PA5 | SD card |
| USART2 | PA2, PA3 | telemetry |
| TIM4 | PB6 | input capture |

3. Build the resource map for **your own project**. One row per resource, one owner each, and mark anything you are unsure about.
4. Explain why `HAL_Delay` inside an interrupt callback can hang forever. Which peripheral does it depend on, and what is the priority condition?

## Part B - Timing budgets

5. Write the formula for the load of an activity from its cost and rate.
6. Compute each load and the total.

| Activity | Cost | Rate | Load % |
|---|---|---|---|
| Tick interrupt | 2 µs | 1000 Hz | |
| ADC read (two channels) | 40 µs | 200 Hz | |
| UART line, 24 chars @ 115 200 | | 10 Hz | |
| LCD line, 16 chars @ 100 kHz | 7.2 ms | 4 Hz | |
| **Total** | | | |

7. Which activity dominates that budget, and by what factor over the next largest? What would you change first, and what would it save?
8. A team wants the LCD refreshed at 30 Hz. Compute the load at 100 kHz and at 400 kHz. Is either acceptable? Give a design that achieves a *visibly* responsive display without either cost.
9. Build the timing budget for **your own project**. State the total, name the dominant item, and say what you would cut first if you had to.

## Part C - Diagnosis

10. For each symptom, name the most likely species and the first thing you would check.

| Symptom | Species | First check |
|---|---|---|
| Display garbles only while the servo sweeps | | |
| Telemetry stops permanently after ~10 minutes | | |
| The servo never moves, from the very first run | | |
| The board resets every few minutes under load | | |
| Readings are correct but arrive in bursts | | |

11. You have six subsystems and one intermittent fault. Describe a bisection plan, and state the worst-case number of steps. Show the reasoning.

12. Give three instruments for observing a system you cannot stop, with the approximate cost and what each reveals.
13. Turn this into a usable bug report: *"it hangs sometimes."* State what you would measure and what a good report would say.
14. Explain how `git bisect` finds the commit that introduced a fault, and what you must be able to do at each step for it to work.

## Part D - Requirements and evidence

15. Rewrite each as a testable requirement.

| Vague | Testable |
|---|---|
| "The logging is reliable" | |
| "The motor responds quickly" | |
| "The sensor is accurate" | |
| "The system is robust" | |

16. Write three testable requirements for **your own project**, each with the test procedure and the result you would accept as a pass.
17. Why does an empty known-issues list read badly at a readiness review? What does a good one look like?

## Part E - The physical layer

18. Explain why two supplies without a common ground produce failures that are hard to diagnose. What is the symptom?
19. A servo stalls and the board resets. Explain the mechanism, and give two fixes — one electrical, one in software.
20. Give two symptoms that a loose jumper and a software race have in common, and one test that distinguishes them.

## Exit self-assessment

☐ I have a complete resource map for my project with no unresolved conflicts.
☐ I have a timing budget with a stated total.
☐ I can name the species of a failure from its symptom.
☐ I can describe a bisection I performed.
☐ My requirements are testable and I know how each is tested.